



Maxim Bejenari

Passport: ... | **Date of birth:** 18/05/2005 | **Place of birth:** Balti, Moldova |

Nationality: Romanian | **Gender:** Male | **Phone number:** (+40) 733259245 (Mobile) |

Email address: maxonbej@gmail.com | **LinkedIn:**

<https://www.linkedin.com/in/bejenari-maxim-8b2746268/> | **github:**

<https://github.com/maxonbejenari> | **Address:** Bacau, Romania (Home)

ABOUT ME

I am a motivated and curious person, always eager to learn and adapt to new challenges. I value collaboration, creativity, and continuous improvement.

WORK EXPERIENCE

INDUSTRIAL ENGINEER INTERN – APERAM MAGNETIC COMPONENTS – 08/2026 – 09/2026 – BALTI, MOLDOVA

During my internship, I actively bridged the gap between theoretical knowledge and industrial application, focusing on mechanical design, software automation, and precision electronic testing. Conducted rigorous precision testing using RLC meters. Successfully diagnosed measurement anomalies by differentiating between physical component defects. Developed and implemented custom Python scripts to automate repetitive mechanical design tasks within CAD environments, significantly reducing design time and minimising the risk of human error.

EDUCATION AND TRAINING

01/09/2021 – 31/05/2024 Balti, Moldova

HIGH SCHOOL DIPLOMA THEORETICAL LYCEUM "LUCIAN BLAGA"

Field of study Mathematics, Physics, Biology, Chemistry

01/10/2024 – CURRENT Bacau, Romania

BACHELOR "Vasile Alecsandri" University of Bacau, Faculty of Engineering

Website <https://www.ub.ro/> | **Field of study** Mechatronics Engineering

01/03/2026 – 01/07/2026 Istria, Croatia

ERASMUS+ ISTRIAN UNIVERSITY OF APPLIED SCIENCES

Website <https://www.iv.hr/en/istrian-university-of-applied-sciences/> | **Field of study** Mechatronics Engineering

LANGUAGE SKILLS

Mother tongue(s): **ROMANIAN**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
RUSSIAN	C1	C1	C1	C1	B2
ENGLISH	B2	B2	B2	B2	B2
GERMAN	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● SKILLS

software engineering | robotics engineering | C++ | Python (computer programming) | Git | Linux | SQL | ROS2 | Electrical Engineering | Mechanical Engineering | CAD | PLC | Arduino | Raspberry Pi | MATLAB | computer vision | OpenCV

● PROJECTS

Mobile App Development Project – Tekwill National Contest, Moldova

Contributed to designing and developing a Java-based mobile application that provides teachers with a digital timetable management tool.

Link <https://github.com/maxonbejenari/E-Orar>

Tekwill National Contest – Algorithm Design (C++)

Participated in a national competition focused on designing and solving algorithms using C++.

01/02/2026 – CURRENT

The Robot Vacuum Cleaner Simulation Prototype

This project is a prototype autonomous vacuum cleaner robot developed entirely in simulation. The robot was designed in Onshape, converted to URDF using onshape-to-robot, simulated in Gazebo, and equipped with:

- SLAM Toolbox for real-time mapping and localization
- NAV2 for autonomous point-to-point navigation
- YOLOv8 for real-time object detection, trained on custom classes (e.g. cube detection)

The goal is to simulate the full software stack of a commercial robot vacuum cleaner — from sensor fusion and mapping to coverage path planning and autonomous navigation.

Link https://github.com/maxonbejenari/Robot_Vacuum_Cleaner_ROS2

Strength Analysis of Pressure Vessels (Erasmus+ Project) | Istrian University of Applied Sciences

- Conducted a detailed strength analysis of spherical pressure vessels under various operating temperatures (up to 200°C), in accordance with the AD 2000-Merkblatt safety standards.
- Calculated the minimum wall thickness and structural safety factor for various industrial steel grades (EN 10025 S355J2, NIOVAL 47).
- Validated the analytical calculations through Finite Element Analysis (FEA) on a 3D model, achieving a precise correlation between the numerical simulations and theoretical results.

Link <https://drive.google.com/file/d/1-iSkkJDhbVIWjR8x-nVtzHDBpjaPaq-p/view?usp=sharing>